

PROGRAM 7 — Deploy Web App using Kubernetes & Minikube

STEP 1: Start Minikube

```
minikube start
```

STEP 2: Verify Minikube status

```
minikube status
```

STEP 3: Create Deployment YAML file

Create file:

```
nano nginx-deployment.yaml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
name: nginx-deployment
```

```
spec:
```

```
replicas: 2
```

```
selector:
```

```
matchLabels:
```

```
app: nginx
```

```
template:
```

```
metadata:
```

```
labels:
```

```
app: nginx
```

```
spec:
```

containers:

- name: nginx

image: nginx:latest

ports:

- containerPort: 80

Save file.

STEP 4: Apply Deployment

```
kubectl apply -f nginx-deployment.yaml
```

STEP 5: Check Deployment & Pods

```
kubectl get deployments
```

```
kubectl get pods
```

STEP 6: Expose Deployment via Service (NodePort)

```
kubectl expose deployment nginx-deployment --type=NodePort  
--port=80
```

STEP 7: Check Services

```
kubectl get svc
```

STEP 8: Access the Web Application

```
minikube service nginx-deployment
```

This opens the Nginx welcome page in your browser.

STEP 9: Cleanup

```
kubectl delete service nginx-deployment
```

kubectl delete deployment nginx-deployment

OUTPUT:

Nginx deployed and accessed via Minikube service URL.

```
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593: ~
Port: 80/TCP
Host Port: 0/TCP
Environment: <none>
Mounts: <none>
Volumes: <none>
Node-Selectors: <none>
Tolerations: <none>
Conditions:
  Type           Status  Reason
  ----           -
  Available       False   MinimumReplicasUnavailable
  Progressing     True    ReplicaSetUpdated
OldReplicaSets: <none>
NewReplicaSet:  nginx-deployment-6f9664446b (2/2 replicas created)
Events:
  Type       Reason              Age   From          Message
  ----       -
  Normal     ScalingReplicaSet   47s   deployment-controller Scaled up replica set nginx-deployment-6f9664446b from 0 to 2
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$ kubectl get services
NAME         TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes   ClusterIP   10.96.0.1    <none>         443/TCP    11h
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$ minikube service nginx-deployment
! Executing "docker container inspect minikube --format={{.State.Status}}" took an unusually long time: 19.005057753s
! Restarting the docker service may improve performance.

✖ Exiting due to GUEST_STATUS: Unable to get control-plane node minikube host status: state: unknown state "minikube": context deadline exceeded

1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$ kubectl expose deployment nginx-deployment \
--type=NodePort \
--port=80
service/nginx-deployment exposed
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$ kubectl get services
NAME         TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes   ClusterIP   10.96.0.1    <none>         443/TCP    11h
nginx-deployment NodePort    10.99.204.106 <none>        80:30205/TCP 9s
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$ minikube service nginx-deployment
! Executing "docker container inspect minikube --format={{.State.Status}}" took an unusually long time: 4.445986873s
! Restarting the docker service may improve performance.
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30205

```
🔗 Opening service default/nginx-deployment in default browser...
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~$
```

